Department of Mathematics
National Institute of Technology Kurukshetra
Discrete Mathematics (MAPC 203)

Time: 08:00 am-08:50 am | Max Marks: 15 | Date: 24/09/25

1. (a) Determine whether each of the following statements is a tautology or a contradiction.
   1. $p \to (q \to r)$
   2. $(p \to q) \to [(p \vee (q \wedge r)) \leftrightarrow q \wedge (p \vee r)]$

(5)

   (b) Rewrite the following statements in the formal form.
   1. Every real number is positive, negative, or zero.
   2. Some real numbers are rational.

   (c) Find the converse of the conditional statement.
   "The home team wins whenever it is raining"

2. (a) What are the negations of the statements
   1. $\forall x(x^2 > x)$.
   2. $\exists(x^2 = 2)$.
   3. $\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \ (0 < |x - a| < \delta \ \Rightarrow \ |f(x) - L| < \varepsilon)$.

(3)

   (b) Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y, where x is $1, 2, or\ 3$ and y is $1, 2, or\ 3$. Write out these prpositional using disjunction and conjunction.
   1. $\forall x \exists y P(x, y)$.

(1)

   (c) Show that the following argument is valid by rule of inference.

$$\begin{array}{c} p \\ p \to q \\ q \to r \\ \hline r \end{array}$$

$q \to p$
$, p$

(d) Write $(p \leftrightarrow q)$ in principle conjunctive normal form. **(2)**

3. (a) Determine whether the following algorithm represents the sum rule or the product rule of counting. Then, explain how it illustrates the corresponding counting principle. Finally, determine the value of k at the end of the execution. **(1)**

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k ← 0
for i_1 = 1 to n_1:
    k ← k + 1
    for i_2 = 1 to n_2:
        k ← k + 1
        ...
        for i_m = 1 to n_m:
            k ← k + 1
```

(b) Three cards are chosen one after the other from a 52-card deck. Find the number m of ways this can be done: (a) with replacement; (b) without replacement. **(3)**